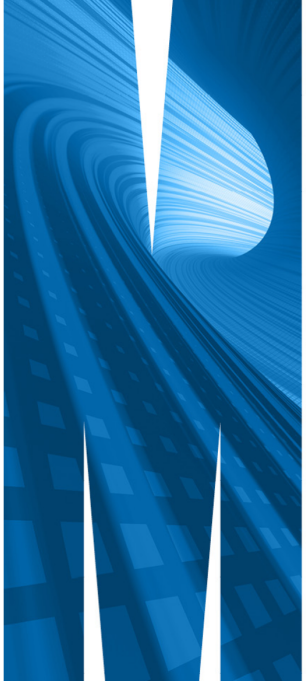
MONASH University

ETX1100/ETX5900
Business Statistics

Week 2: Probability and distributions

1 December 2026



Outline

1 Events and probability rules

2 Two-way categorical tables

3 Continuous probability models

Today’s question

How likely is a business event, and which other information changes that probability?

We move from last week’s frequencies to events, rules, two-way tables and conditional probabilities. We then use a smooth probability model for a continuous measurement.

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Probability turns a count into a chance

ETF defines probability as a **numerical value representing the chance, likelihood or possibility that an event will occur**. It lies between 0 and 1.

An **event** is one or more outcomes of an experiment. A **sample space** S collects all possible simple outcomes. For one fair die roll, $S=\{1,2,3,4,5,6\}$ and $P(S)=1$.

A sample frequency can estimate a probability when the sample and observation process are suitable. Always name the population or process the estimate describes.

AND, OR, NOT

Language	Notation	Meaning
A **and** B	$A \cap B$	Both events occur
A **or** B	$A \cup B$	At least one occurs, including both
Not A	A^c	A does not occur

The addition rule removes the double-counted intersection:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

If A and B cannot occur together, they are **mutually exclusive**

The intersection is counted once

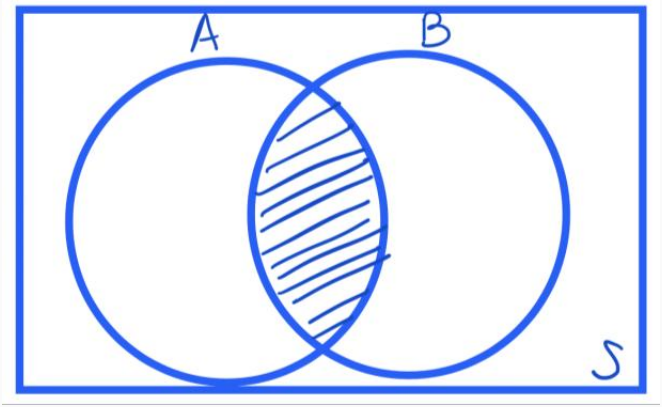


Figure 1: Intersection of two events from ETF's probability lecture.

The shaded overlap is $A \cap B$. When adding `P(A)` and `P(B)`, that

Watch me: a die roll

Let `A` be “even” and `B` be “less than 5.” From the six equally likely outcomes:

- `A={2,4,6}` and `B={1,2,3,4}`.
- The intersection is `{2,4}`.
- $P(A \cup B) = 3/6 + 4/6 - 2/6 = 5/6$.

The same result comes from listing `{1,2,3,4,6}`. ETF uses this check to distinguish **or** from **and**.

Now you try: translate the event

Time allowed: 5 minutes

For a randomly selected business invoice, let A be “paid late” and B be “contains an error.” Write in words: $A \cap B$, $A \cup B$ and `A^c`. If `P(A)=0.12`, `P(B)=0.08` and $P(A \cap B) = 0.03$, calculate the union. Explain the subtraction.

Mutually exclusive and independent

These are different ideas:

- **Mutually exclusive:** the events cannot happen together.
- **Independent:** knowing one occurred does not change the probability of the other.

If two nonzero-probability events are mutually exclusive, observing one rules out the other. They are therefore not independent. We test independence through conditional probabilities after building a table.

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From one variable to two

A **contingency table** counts observations across two categorical variables. ETF Seminar Week 3 extends a one-way PivotTable into a two-way table.

For the Week 2 Marketing workbook, place `Have Tried` in rows and `Income` in columns. Count `Person ID` in Values. The row totals, column totals and grand total determine which denominator belongs to each question.

Watch me: build the Marketing table

1. Select the `Marketing` table and choose **Insert > PivotTable**.
2. Drag `Have Tried` to **Rows**, `Income` to **Columns**, and `Person ID` to **Values** as **Count**.
3. Make a copy showing **% of Grand Total** for joint and marginal probabilities.
4. Make a second copy showing **% of Row Total** to condition on `Have Tried`.
5. Check one cell with `COUNTIFS` and one marginal count with `COUNTIF`.

A table percentage without its denominator is not yet an

Joint and marginal probabilities

A **joint probability** describes a pair of categories, the intersecting cell divided by the grand total. A **marginal probability** describes one variable, the relevant row or column total divided by the grand total.

ETF's workshop asks for both "has not tried" and "earns above AUD 50,000 **and** has not tried." The first is marginal; the second is joint. The words in the question point to different cells.

Now you try: read the correct cell

Time allowed: 7 minutes

Using your PivotTable, estimate (1) *P*(not tried) and (2) *P*(income above 50,000 and not tried). State the numerator and the **all-respondent** denominator for each. Check one result with an Excel count formula.

Conditional probability changes the denominator

ETF describes conditional probability as the probability of one event **given information that another event has occurred**:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{\text{count of both A and B}}{\text{count of B}}, \qquad P(B) > 0.$$

The condition, after the vertical bar, defines the smaller group. $P(A \mid B)$ generally differs from $P(B \mid A)$.

Watch me: condition on “not tried”

ETF Workshop Week 3 asks: among respondents who **have not tried** the product, what proportion earn above AUD 50,000?

Use the count of “not tried **and** above 50,000” as numerator. Divide by the **not-tried row total**. In a PivotTable, `% of Row Total` gives this when `Have Tried` is in rows. Say the denominator aloud before reading the percentage.

Now you try: reverse the condition

Time allowed: 6 minutes

Find the proportion who **have not tried** the product among respondents earning above AUD 50,000. Change the denominator, not just the wording. Explain why this result need not equal the previous conditional probability.

Union probabilities from a table

For A = “has tried” and B = “lives alone,” add the two marginal counts and subtract their overlapping cell once. Divide the result by the grand total.

The same answer can be checked by counting every respondent who meets **A or B**. This is particularly useful when a business question combines two warning indicators.

Independence: compare conditional and marginal

Events A and B are independent if learning B does not change A's probability:

$$P(A \mid B) = P(A), \quad \text{equivalently} \quad P(A \cap B) = P(A)P(B).$$

Compare the conditional "tried" rates across income categories with the overall "tried" rate. A sample difference suggests association; it does not establish causation or by itself constitute a formal test.

Now you try: one paragraph of evidence

Time allowed: 8 minutes

Choose two categories in `Marketing`. Report a joint probability, a conditional probability and an independence comparison. Name each denominator and finish with one cautious business sentence.

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A histogram is observed data; a model is an assumption

A histogram groups the measurements we actually saw. ETF Seminar Week 4 then introduces a smooth **probability density** `f(x)` to represent a proposed distribution of future or population values.

Area under the curve is probability. The total area is 1. A model can be useful for risk calculations, but first check whether its shape is credible for the business variable.

The Normal distribution

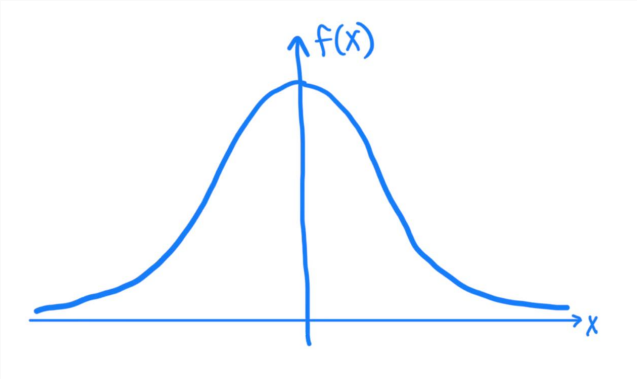


Figure 2: ETF's smooth density schematic used to introduce continuous probability.

A Normal distribution is symmetric and bell-shaped, described

Excel functions answer three different questions

Business question	Excel expression when $X \sim$
Below x?	`=NORM.DIST(x,mean,sd,T`
Above x?	`=1-NORM.DIST(x,mean,sd`
Which x has cumulative probability q?	`=NORM.INV(q,mean,sd)`

The fourth argument `TRUE` requests a cumulative probability. For a continuous variable, “less than” and “less than or equal to” have the same probability.

Watch me: a service threshold

ETF Seminar Week 4 uses $X \sim N(10, 3)$ to illustrate the Excel function: `P(X<5)=NORM.DIST(5,10,3,TRUE)`. The complement gives `P(X>5)`.

Before pressing Enter, sketch which tail is needed. Check that a probability is between 0 and 1 and that a threshold below the mean has cumulative probability below 0.5.

Now you try: blood-pressure risk

Time allowed: 8 minutes

ETF Workshop Week 4 assumes adult systolic blood pressure is Normal with mean 124 mmHg and SD 10 mmHg. In Excel, find the probabilities **below 117** and **above 140**, then the reading that marks the highest 15%. State each event before choosing `NORM.DIST` or `NORM.INV`.

Standard Normal and Student's t

A **standard Normal** has mean 0 and SD 1. Standardisation puts unlike measurements on a common scale:

$$z = \frac{x - \mu}{\sigma}.$$

ETF introduces the **t distribution** because estimating an unknown population SD from a sample adds uncertainty. It resembles the standard Normal but has heavier tails for small degrees of freedom; as degrees of freedom grow, the two become close. Week 3 uses these critical values for intervals.

Check the model before the calculation

The ETF price example is right-skewed. A symmetric Normal curve may be a poor model for those raw prices. Use a stated Normal assumption when a problem gives it; otherwise inspect the histogram, possible outliers, units and sampling context.

A precise Excel probability from an unsuitable model can be misleading.

Closing chain

Business event $\rightarrow$ sample space or model $\rightarrow$ denominator or tail $\rightarrow$ Excel calculation $\rightarrow$ validation $\rightarrow$ plain-language interpretation.

Next week we ask how much a probability or mean estimated from one sample might change with another sample.